OLDA5302M Summative Assessment 1: Project planning form

Project description

This project will address the *policing rural communities project brief,* focussing on open source street-level crime data (Home Office, 2025).

Section 1: Subject knowledge

Describe the main aim of your project.

The project aims to analyse crime patterns and outcomes between rural and urban areas in Kent. It will apply spatial enrichment and machine learning techniques to evaluate if observed patterns from 2024-2025 data align with previously identified community priorities (Office of the Kent Police and Crime Commissioner, 2021).

What are your preliminary research questions?

I. How do the frequency, types, and outcomes of crime differ between rural and urban areas in Kent?
II. Can crime type be predicted using spatial and temporal features?
III. Do predicted crime outcomes differ systematically between rural and urban areas, and do these patterns align with community policing priorities?

Section 2: Data knowledge

How is your research question related to the available data?

The research questions are directly supported by fields within UK street-level crime data published by the Home Office (2025); temporal, crime outcome and type features support both exploratory analysis and predictive modelling. To distinguish rural and urban location, the dataset will be enriched using Office for National Statistics (2021) rural–urban classifications linked through LSOA identifiers. This enrichment supports both comparative analysis and classification tasks. Additionally, rural criminology literature by Bullock & Garland (2023) highlights perceptions of under-prioritisation in rural communities, supporting investigation into whether crime outcomes align with community policing priorities.

Section 3: Data preparation

What steps are required to prepare the data and determine whether your project is feasible?

To satisfy all preliminary research questions, datasets from the Home Office (2025) and the Office for National Statistics (2021) will be joined to classify crime locations as rural or urban. This will be followed by standard preprocessing including the removal of null and duplicate data points. Feature engineering steps will also be required, such as grouping temporal variables into appropriate time buckets, and categorical encoding of crime type and outcome to use as features for classification modelling. However, insights from exploratory analysis may alter this approach in the presence of autocorrelation, class imbalance or low feature relevance. This could potentially influence model selection and/or revisions of the preliminary research questions.

Section 4: Data modelling

Data Science technique	Reason for use	Module code
Multinomial Logistic Regression	This technique will be used to predict crime type, a multiclass target. The probabilistic outputs support analysis of spatial and temporal influence on crime.	OMAT5201M
Random Forest / Gradient Boosting	This technique will be used to classify crime outcome based on categorical, temporal and spatial features which are likely to exhibit complex non-linear relationships. Gradient boosting methods may be used to enhance predictive performance in the presence of class imbalance.	OMAT5200M

Section 5: Project evaluation

How will you evaluate your model?

All models will be evaluated using train–test splits and cross-validation to assess reliability, optimise hyperparameters, and reduce overfitting risk. Multinomial logistic regression performance will be assessed using F1-score and AIC to evaluate prediction and model fit. The influence of spatial and temporal features will be interpreted with model coefficients and odds ratios. Evaluation of ensemble methods will be informed by potential adaptations from EDA. Confusion matrices, precision-recall, and F1-scores will be compared across rural and urban subsets to assess generalisability, while ROC-AUC may be used if the model fit favours a binary crime outcome.

Section 6: Reflection

Do you anticipate any potential challenges or obstacles? Briefly explain how you would plan to deal with these.

Following exploratory analysis there is potential for class imbalance due to spatial or categorical bias. Temporal autocorrelation may also affect model reliability and independence assumptions. Should any of these issues occur, the cardinality of categorical and temporal features may be reduced. Undersampling methods for spatial features may also be considered. The prediction of crime outcomes may require cardinality reduction based on the level of noise, imbalance and interpretability of a multi-class ensemble model.

Are there any ethical considerations associated with your project?

Crime outcomes contain ambiguous categories such as "*investigation complete; no suspect identified*", which may require cardinality reduction to a binary class for modelling. This category could be identified as either resolved or unresolved crime due to the complexity of criminal investigation, however this should not be interpreted as evidence of true criminal activity or policing effectiveness. Additionally, the data may exhibit spatial bias which creates the risk of misrepresenting rural communities. Recorded crime reflects police reporting and enforcement behaviour rather than true crime incidence, requiring careful interpretation when discussing spatial fairness.

Have you incorporated a source of feedback into your project?

This is an individual project; however, feedback has been incorporated through informal peer discussion and weekly webinar Q&A sessions. This has helped provide clarification on the scope of the project, particularly in refining the research questions to ensure the modelling approach remained feasible within the module timeframe.

Section 7: Additional information

References

Home Office. 2025. *Kent Police street-level crime & outcome data*. [Online]. [Accessed 19 May 2026]. Available from: https://data.police.uk/data

Office for National Statistics (ONS). 2021. *Rural Urban Classification (2021) of LSOAs in EW*. [Online]. [Accessed 20 May 2026]. Available from: https://geoportal.statistics.gov.uk/datasets/ons::rural-urban-classification-2021-of-lsoas-in-ew/explore

Office of the Kent Police and Crime Commissioner, 2021. *Police and Crime Plan Survey*. [Online]. [Accessed 19 May 2026]. Available from: https://www.kent-pcc.gov.uk/SysSiteAssets/media/downloads/get-in-touch/previous-consultations/police-and-crime-plan-survey-report-2021.pdf

Bullock, K. & Garland, J., 2023 Policing Rural Crimes and Rural Communities in England. [Online]. *International Journal of Rural Criminology.* **8**(1), pp.41-58. Available from: https://doi.org/10.18061/ijrc.v8i1.9561